Section 2–6 Signed Numbers:

23. Express each decimal number in binary as an 8-bit sign-magnitude number:

(b) – 85

24. Express each decimal number as an 8-bit number in the 1's complement form:

(c) – 99

25. Express each decimal number as an 8-bit number in the 2's complement form:

(b) – 68

Section 2–6 Signed Numbers:

29. Express each of the following sign-magnitude binary numbers in single-precision floating point format:

(b) 100110000011000

30. Determine the values of the following single-precision floating-point numbers:

(b) 0 11001100 10000111110100100000000

Section 2–7 Arithmetic Operations with Signed Numbers:

33. Perform each addition in the 2's complement form:

(b) 11011001 + 11100111

34. Perform each subtraction in the 2's complement form:

(b) 01100101 – 11101000

36. Divide 10001000 by 00100010 in the 2's complement form.

Section 2–10 Binary Coded Decimal (BCD):

49. Convert the following decimal numbers to BCD:

(h) 547

51. Convert each of the BCD numbers to decimal:

(h) 0001011010000011

53. Add the following BCD numbers:

(e) 00100101 + 00100111

Section 2–11 Digital Codes:

56. Convert each binary number to Gray code:

(b) 1001010

57. Convert each Gray code to binary:

(b) 00010

Section 2–12 Error Codes:

64. Determine which of the following odd parity codes are in error:

(a) 11110110

65. Attach the proper even parity bit to each of the following bytes of data:

(a) 10100100

$\boxed{23}$ (b) $-85_{10} = ?$ (sign-magnitude)

$85_{10} = 1010101_2$

$-85_{10} = \underset{\text{Sign}}{1}\ \underbrace{1010101}_{\text{magnitude}}$

$\boxed{24}$ (c) $-99_{10} = ?$ (1's comp.)

$99_{10} = 1100011_2$

$-99_{10} =$ 1's comp. of $\overbrace{01100011}^{8\text{-bit}}$

$= 10011100$

$\boxed{25}$ (b) $-68_{10} = ?$ (2's Comp.)

$68_{10} = 1000100$

$-68_{10} =$ 2's comp. of $\overbrace{01000100}^{8\text{-bit}}$

$= 10111100$

$\boxed{29}$ (b) $1001100000011000_{SM} = ?$ (single-precision)

$\Downarrow$

$-11000001100 0_2 = -1.1000011 * 2^{1011}$

$S = 1,\quad E = 01111111 + 1011 = 10001010,\quad F = 10000011 00\ldots$

Number $= 1\ \overbrace{10001010}^{8}\ \overbrace{10000011 0000 0000 0000 000}^{23}$

$\boxed{30}$ (b) $0\ 11001100\ 10000111110100100000000 = ?_{10}$

$(11001100 - 01111111)$

Numer $= (-1)^0 (1 + 0.10000111110100 1) * 2$

$= +1.10000111110100 1 * 2^{1001101}$

33 (b)

$$
\begin{array}{rl}
11011001 & \longrightarrow -00100111 \rightarrow -39_{10} \\
+\ 11100111 & \longrightarrow -00011001 \rightarrow -25_{10} \\
\hline
\boxed{\cancel{1}}\ 11000000 & \longrightarrow -01000000 \rightarrow -64_{10}
\end{array}
$$

---

36 $\underbrace{10001000}_{-120_{10}} \div \underbrace{00100010}_{34_{10}} = ?$

Dividend is -ve, so we need to calculate the positive number that has the same magnitude, i.e. take its 2's complement $\Rightarrow$ 01111000

$$
\begin{array}{r|l}
 & 00000011 \leftarrow Q \\
\hline
00100010 & 01111000 \\
 & 00100010 \\
\hline
 & 00110100 \\
 & 00100010 \\
\hline
 & 00010010 \leftarrow R
\end{array}
$$

Since Dividend is -ve & Divisor is +ve

Quotient $= -Q =$ 2's comp. of (00000011)

$= 11111101 \leftarrow -3_{10}$

Remainder $= -R =$ 2's comp. of (00010010)

$= 11101110 \leftarrow -18_{10}$

---

34 (b)

$$
\begin{array}{rl}
01100101 & \longrightarrow +101_{10} \\
-\ 11101000 & \longrightarrow -00011000 \rightarrow -24 \\
\hline
01100101 & \\
+\ 00011000 & \\
\hline
01111101 & \longrightarrow +125_{10}
\end{array}
$$

49 (h) $547_{10} = ?_{BCD}$

5 → 0101, 4 → 0100, 7 → 0111

---

51 (h) $0001\,0110\,0000\,0011_{BCD} = ?_{10}$

1 6 8 3

---

53 (e)

$$\begin{array}{r} 0010\ 0101 \\ +\ 0010\ 0111 \\ \hline 0100\ 1100 \\ +\ \ \ \ \ \ \ 0110 \\ \hline 0101\ 0010 \end{array}$$

> 1001

Correction

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56 (b) $1001010_2 = ?_{Gray}$

1 ⊕ 0 ⊕ 0 ⊕ 1 ⊕ 0 ⊕ 1 ⊕ 0

1 1 0 1 1 1 1

---

57 (b) $00010_{Gray} = ?_2$

0 ⊕ 0 ⊕ 0 ⊕ 1 ⊕ 0

0 0 0 1 1

$\boxed{64}$ (a) $1111\ 0110_2$ ✓ or X ?

# of 1's = 6 ← even

since odd parity is used ⟹ there is an error!

$\boxed{65}$ (a) $\boxed{?}1010\ 0100_2$ even parity

# of 1's = 3 ← odd

since even parity is used ⟹ parity bit = 1

Answer : $\boxed{1}1010\ 0100$